WORKSHEET

Classifying variables 2

1 Classify the following variables as discrete or continuous.

a the length of a field

b the time it takes to learn how to classify variables

c the size of a shoe

d the mass of a mountain

e the number of people living in a house

f the number of heads recorded after tossing a coin

g the amount of rain recorded during a storm

h the number of points scored in a basketball game

i the price of a smartphone

j the distance from home to school

k the number of cars waiting in a queue to cross an intersection

l the time a car waits in a queue to cross an intersection.

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2 Determine if the following are random variables. For those that are random variables, state if they are discrete or continuous.

a the number of heads recorded after tossing a coin

b the number of hearts in a 5 card hand drawn from a well-shuffled deck

c the colour of a randomly chosen card from a well-shuffled deck of cards

d the lifetime of a light bulb

e the manufacturer of a randomly chosen car

f the height of a randomly chosen student

g the name of a randomly selected Australian city

h the brand of mobile phone owned by the next student to enter the classroom

i the number of houses in a randomly chosen street

j the size of a randomly chosen block of land.

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Answers

1 **a** continuous
b continuous
c discrete
d continuous
e discrete
f discrete
g continuous
h discrete
i discrete
j continuous
k discrete
l continuous

2 **a** yes, discrete
b yes, discrete
c no, not numerical
d yes, continuous
e no, not numerical
f yes, continuous
g no, not numerical
h no, not numerical
i yes, discrete
j yes, continuous

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